

Q1. Write a note on Cyber Security

Cyber Security refers to the practice of protecting computers, networks, servers, mobile devices, software, and data from unauthorized access, cyber attacks, damage, theft, or disruption.

The main purpose of cyber security is to ensure that information remains **confidential, accurate, and available** to authorized users.

Importance of Cyber Security

1. **Protects personal data** such as passwords, bank details, and identity information.
2. **Prevents unauthorized access** to computers and networks.
3. **Protects organizations** from data breaches and financial losses.
4. **Prevents malware and virus attacks.**
5. **Maintains privacy** of users and organizations.
6. **Ensures business continuity** by protecting systems from disruption.

Common Cyber Security Measures

- Using strong passwords and multi-factor authentication.
- Installing antivirus and anti-malware software.
- Keeping operating systems and applications updated.
- Using firewalls.
- Encrypting sensitive information.
- Regularly backing up important data.
- Educating users about phishing and other cyber threats.

Conclusion: Cyber security is essential in today's digital world because it protects information, systems, and users from continuously evolving cyber threats.

Q2. Explain CIA Triad of Information Security

The **CIA Triad** is a fundamental model of information security. CIA stands for:

C – Confidentiality

I – Integrity

A – Availability

1. Confidentiality

Confidentiality means ensuring that information is accessible **only to authorized users**.

Example: A bank account password should only be accessible to the account holder.

Methods:

- Passwords
- Encryption
- Access control
- Multi-factor authentication

2. Integrity

Integrity means ensuring that information remains **accurate, complete, and unaltered** by unauthorized people.

Example: A student's marks stored in a college database should not be changed without authorization.

Methods:

- Hashing
- Digital signatures
- Access controls
- Data validation

3. Availability

Availability means ensuring that information and systems are **accessible to authorized users whenever required**.

Example: An online banking website should be available when customers need to access their accounts.

Methods:

- Regular backups
- Disaster recovery
- Redundant systems
- Protection against DDoS attacks

CIA Triad Example

Principle	Meaning	Example
Confidentiality	Prevent unauthorized access	Password protection
Integrity	Prevent unauthorized modification	Hashing

Availability

Keep systems accessible

Backup servers

Conclusion: The CIA Triad provides the basic foundation for designing and maintaining secure information systems.

Q3. What are the different types of Cyber Crimes?

Cyber crime is any illegal activity in which a computer, network, digital device, or the internet is used as a tool or target.

Major Types of Cyber Crimes

1. **Hacking**
Unauthorized access to a computer system or network.
2. **Phishing**
Attackers send fake emails, messages, or websites to trick users into revealing passwords, banking information, or other sensitive data.
3. **Identity Theft**
Stealing someone's personal information and using it for fraudulent activities.
4. **Online Banking and Financial Fraud**
Unauthorized transactions or theft of money through online banking, cards, or digital payment systems.
5. **Cyber Stalking**
Repeatedly monitoring, threatening, or harassing a person using digital platforms.
6. **Cyber Bullying**
Using social media, messaging platforms, or other online services to threaten, insult, or harass someone.
7. **Ransomware Attacks**
Malware encrypts a victim's files and demands money in exchange for restoring access.
8. **Data Theft**
Unauthorized stealing or copying of confidential information from individuals or organizations.
9. **Denial-of-Service (DoS/DDoS) Attacks**
Overloading a server or network with traffic so that legitimate users cannot access the service.
10. **Email Spoofing**
Creating a fake email identity or sender address to deceive the recipient.
11. **Cyber Extortion**
Threatening a person or organization with releasing or damaging information unless a demand is fulfilled.

12. **Software Piracy**

Illegal copying, distribution, or use of copyrighted software.

Conclusion: Cyber crimes can cause financial losses, privacy violations, reputational damage, and disruption of services. Awareness and proper security practices are important to prevent them.

Q4. What are the different types of Malware?

Malware stands for **Malicious Software**. It is software intentionally designed to damage systems, steal information, gain unauthorized access, or disrupt computer operations.

Different Types of Malware

1. **Virus**
A virus attaches itself to a legitimate file or program and spreads when the infected file is executed.
2. **Worm**
A worm can replicate itself and spread automatically across networks without requiring a user to execute an infected file.
3. **Trojan Horse**
A Trojan disguises itself as legitimate or useful software but performs malicious activities after installation.
4. **Ransomware**
Ransomware encrypts files or locks systems and demands payment from the victim.
5. **Spyware**
Spyware secretly monitors user activities and may collect sensitive information such as passwords and browsing data.
6. **Adware**
Adware displays unwanted advertisements and may sometimes track user activity.
7. **Rootkit**
A rootkit is designed to hide malicious software or unauthorized activities deep inside a system.
8. **Keylogger**
A keylogger records keystrokes made by a user and can be used to steal passwords and other sensitive information.
9. **Bot/Botnet Malware**
Malware can turn infected devices into remotely controlled **bots**. A collection of such devices is called a **botnet**, which can be used for activities such as DDoS attacks.
10. **Fileless Malware**
Fileless malware operates primarily through legitimate system tools and memory

rather than relying on a traditional malicious executable file, making detection more difficult.

Difference between Common Malware

Malware	Main Characteristic
Virus	Attaches to files and spreads when executed
Worm	Self-replicates across networks
Trojan	Pretends to be legitimate software
Ransomware	Encrypts/locks data and demands payment
Spyware	Secretly monitors and collects information
Adware	Displays unwanted advertisements
Rootkit	Hides malicious activity
Keylogger	Records keystrokes
Botnet malware	Creates remotely controlled infected devices
Fileless malware	Operates mainly through memory/legitimate tools

Conclusion: Malware is one of the major threats to computer systems. Using updated software, antivirus protection, strong security practices, and avoiding suspicious links and files can help reduce malware infections.